

|   |     |       |                                                                                                                                                                                                                                                                 |                |
|---|-----|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| 2 | (a) | (i)   | product of mass and (linear) velocity                                                                                                                                                                                                                           | B1             |
|   |     | (ii)  | force equal <u>rate of change of momentum</u> <b>or</b> <u>change in momentum per unit time</u>                                                                                                                                                                 | B1             |
|   | (b) | (i)   | maximum pressure = $55 \times 10^6$ Pa<br>maximum force = $55 \times 10^6 \times 2.8 \times 10^{-4} = 15400$ N<br>maximum acceleration = $15400 / 0.300 = 5.13 \times 10^4 \text{ m s}^{-2}$                                                                    | C1<br>A1       |
|   |     | (ii)  | momentum change = area under graph $\times$ cross-sectional area of barrel<br>evidence of correct estimation of area under $p$ - $t$ graph<br>$18 \text{ kg m s}^{-1} \leq \Delta p \leq 20 \text{ kg m s}^{-1}$                                                | M1<br>M1<br>A1 |
|   |     | (iii) | <i>either</i> excess pressure or force > original, so larger momentum change or impulse<br><i>or</i> excess pressure or force > original, so larger (average) acceleration<br><i>or</i> force > original, larger work done so larger increase in kinetic energy | M1             |

2

|  |  |  |                                        |    |
|--|--|--|----------------------------------------|----|
|  |  |  | final speed of projectile $>$ original | A1 |
|--|--|--|----------------------------------------|----|

|   |     |     |                                                                                     |    |
|---|-----|-----|-------------------------------------------------------------------------------------|----|
| 2 | (a) | (i) | (constant speed, $ca$ ) acceleration (is always) normal / perpendicular to velocity | B1 |
|---|-----|-----|-------------------------------------------------------------------------------------|----|